

Shuai Ma

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Education

Seattle University

Seattle, WA, United States

Albers School of Business & Economics

M.S. in Business Analytics, STEM-designated

2018–2020

- Relevant coursework: programming for business analytics, applied econometrics, statistical learning, forecasting, big-data analytics, data management, data visualization, and data architecture.
- Capstone: *Influential Factors on Residential Property Prices Using Data-Mining Methods*; analyzed Pierce County housing transactions using statistical and machine-learning methods.

University of Cincinnati

Cincinnati, OH, United States

College of Design, Architecture, Art, and Planning

M.S. in Architecture, research-based program with an Urban Development focus

2009–2011

- Research training: urban theory, research methods, pre-thesis research, social-science research for planners, land-use control, international development policy, and thesis-based scholarly writing.
- Thesis: *Impact of High-Speed Rail Station Development on Surrounding Urban Form: The Beijing–Tianjin Case*.

Hebei University of Technology

Tianjin, China

Project 211 and Double First-Class university

B.Eng. in Urban Planning and Design

2003–2008

- Undergraduate planning foundation: planning studios, regional planning, urban design, urban economics, urban geography, transportation planning, urban ecology, planning code, and spatial information technology.
- NACES course-by-course evaluation: equivalent to a U.S. bachelor's degree in Urban and Regional Planning.

Research Interests

Housing capitalization of flood risk and mitigation investment; distributional and environmental-justice effects of climate-adaptation policy in flood-prone communities; land-use regulation and real estate markets; causal inference and urban informatics, including GIS and explainable machine learning, for policy evaluation.

Research Papers and Writing Samples

Hazard or Amenity? Flood Risk, Waterway Proximity, and Neighborhood Inequality in Housing Prices 2025–Present

Sole-authored PhD writing sample; empirical study of Pierce County, Washington

- Examines how flood risk, waterway proximity, and neighborhood inequality are capitalized into housing prices using 47,763 single-family sales linked to parcel, FEMA flood-risk, and neighborhood data.
- Estimates hedonic regression models with tract-clustered standard errors and conducts spatial analysis to evaluate whether flood-risk effects are bundled with waterfront amenity.
- Uses flood-income-waterway typologies, machine-learning models, and SHAP interpretation to motivate a future causal research agenda on flood-mitigation investment and distributional effects.

Strategic Management's Foundational Silence

2025–Present

First-author coauthored manuscript in revision with Rubina Mahsud and Greg Prussia; developed from a 2020 Strategic Management Society conference proposal

- Built and cleaned a corpus of 3,202 Strategic Management Journal articles and applied classical natural language processing (NLP) methods, local open-source large language models (LLMs), and cloud-based LLMs to analyze the field's dominant vocabulary.
- Develops the concept of “foundational silence” to argue that poverty, war, disease, hunger, homelessness, and related human concerns should be treated as central social contexts rather than peripheral externalities.

Conference Presentations

Strategic Management Society 2060: From Effectiveness to Impact, from Success to Significance

October 2020

Strategic Management Society Annual Conference, London, United Kingdom

Mahsud, R., and Ma, S. Presented by Shuai Ma.

Selected Analytical Projects

Sponge City and Low-Impact Development Technical Research and Application

2017

Research leader; led a two-person, three-month applied research effort on stormwater adaptation

- Synthesized sponge-city policies, Low Impact Development (LID) technologies, and multi-city case materials into technical guidance for runoff control, rainwater reuse, water-quality treatment, and urban flood-risk reduction.
- Evaluated permeable paving, green roofs, bioswales, rain gardens, wetlands, detention facilities, and runoff-control scenarios using catchment analysis, water-balance calculations, and stormwater simulation outputs.

Empirical Analysis of Guangzhou Residential Rental Markets

2021

Sole-authored applied econometric project using scraped public rental-listing data

- Collected and cleaned approximately 3,000 publicly available online rental listings and modeled Guangzhou rent variation using structural housing attributes, rental type, floor information, and district-level location indicators.
- Applied OLS regression, stepwise variable selection, VIF/kappa multicollinearity diagnostics, PCA, and exploratory clustering to evaluate key rental-price determinants.

Serviced Apartment and Long-Term Rental Housing Sector Studies

2016; 2021

Sole-authored mixed-methods housing-market research using field visits, web-scraped data, R, and Python

- Conducted field visits to dozens of serviced-apartment projects and compiled public online listing data to compare rents, occupancy, tenant profiles, product layouts, amenities, operators, and location patterns.
- Analyzed 987 Guangzhou serviced-apartment listings using descriptive statistics, cross-tabulation, chi-square tests, and tree-based feature-importance models to evaluate brand, district, accessibility, and unit-level rent drivers.

Comparative Policy Study of New Industrial Land (M0) in the Greater Bay Area

2021

Sole-authored study of land-use regulation, industrial redevelopment, and zoning-like reform

- Compared M0 land-use policies across Shenzhen, Guangzhou, Dongguan, Foshan, Huizhou, Zhongshan, and Zhuhai, examining industry eligibility, site-selection rules, FAR requirements, subdivision and transfer restrictions, land tenure, and performance monitoring.
- Synthesized national, provincial, and municipal policy documents to evaluate how regulatory design supports industrial upgrading while limiting speculative real-estate conversion of industrial land.

Highest and Best Use (HBU) Optimization for Business-Park Development

2021

Initiated and led internal real estate analytics study using market research, regulatory constraints, and Python linear programming

- Surveyed 11 competing business-park projects in Foshan to parameterize 16 candidate product typologies, including multi-floor manufacturing buildings, detached warehouses, and independent offices.
- Formulated a Python PuLP optimization model to maximize gross development value under FAR, building-height, plot-coverage, and green-space-ratio constraints.
- Tested alternative product-mix scenarios to compare value maximization with market absorption, translating HBU analysis into a data-driven development strategy.

Jiangsu Provincial Low-Carbon Urban System Strategy

2011

Core ARUP project-team member; early-career foundation in regional carbon accounting and low-carbon planning

- Contributed to an IPCC-based carbon-audit framework linking urban-system planning with activity-data and emissions-factor methods across land use, buildings, industry, transportation, water, waste, ecological space, and alternative energy.
- Supported province-scale baseline emissions estimation for Jiangsu's urban system, translating carbon-accounting results into planning-relevant evidence for low-carbon strategy and policy design.

Technical Skills

Programming and Data Analysis: Python (pandas, NumPy, GeoPandas, statsmodels, scikit-learn, SHAP, PuLP, Matplotlib, web scraping); R (tidyverse, ggplot2/GGally, R Markdown); SQL.

Statistical and Econometric Methods: hedonic price modeling, OLS regression with clustered standard errors, model selection and diagnostics, PCA, clustering, and scenario analysis.

Machine Learning and Model Interpretation: tree-based models (decision trees, random forests), feature-importance analysis, and SHAP-based model interpretation.

Text and AI-Based Analysis: natural language processing (NLP), keyword-extraction methods, corpus construction, and large language models (LLMs), including open-source and cloud-based models.

GIS, Spatial, and Environmental Analysis: QGIS; spatial data processing, floodplain mapping, spatial autocorrelation, catchment and water-balance analysis, stormwater simulation outputs, and IPCC-based carbon accounting.

Research Workflow and Reporting: Quarto, LaTeX, Zotero (Better BibTeX), Git/GitHub, and Microsoft Excel.

Professional Experience

Founder & Principal Researcher, Beamlinked Consulting Co., Ltd.

2023–Present

- Founded an independent consulting and research practice focused on urban development, real estate markets, sustainable planning, and policy-oriented analytics.
- Leads applied research and advisory work on business-park development, housing and real estate markets, land-use policy, and climate-related planning issues.
- Develops AI-assisted research workflows, including diffusion-model applications for spatial-design exploration and customized language-model tools for sustainability disclosure review.

Vice Director, Property Asset Management, Ping An Group

2020–2023

- Directed planning and design review across the real estate portfolio, including new-town projects and business parks, and contributed to company-wide planning and design guidelines.
- Served on the internal research think tank, producing regular analytical reports for senior management on real estate markets, urban policy, infrastructure investment, and industrial-park planning.
- Participated as a core ESG committee member, reviewing project sustainability performance and supporting LEED v4.1 O+M Platinum certification work for the Ping An International Finance Center.
- Delivered internal professional training as a company-certified lecturer, communicating planning, real estate, and sustainability knowledge to professional audiences.

Director, Planning and Development, China Fortune Land Development

2017–2018

- Directed town planning, urban design, land-use planning, statutory-planning coordination, and water-green-system planning for two public-private-partnership development districts of roughly 3,200 acres each.
- Co-developed a group-wide green-development guideline adopted as a standard for new real estate development projects.

Project Manager, AtkinsRéalis / Atkins

2013–2017

- Led planning and design teams across town planning, master planning, business and industrial parks, commercial complexes, tourism districts, and sustainability-oriented urban development projects.
- Directed applied research on microclimate analysis, building physics, environmental simulation, and performance-based design methods, including CFD, PMV, whole-building energy simulation, and radiation analysis.
- Served as an internal sustainability consultant, helping project teams apply LEED, BREEAM, WELL, China Green Building Label, EEWB, and CASBEE standards.

Urban Planner, Arup

2011–2013

- Contributed as a project-team member to low-carbon planning, urban design, waterfront development, transit-oriented development, and regional planning projects.
- Supported carbon-audit, green-building economics, low-carbon ecological planning, and cross-scalar planning research for department-level studies and publications.
- Supported a National Basic Research Program study and contributed research materials later incorporated into two published department-led books: *Low-Carbon Ecological Space: Rethinking Cross-Dimensional Planning Methods* and *Study on the Economics of Green Buildings in China*.

Assistant Planner, Tongji University Urban Planning and Design Institute

2007–2009

- Supported regional planning, comprehensive urban development planning, tourism planning, historic-district design, and industrial-park planning projects.

Certifications and Professional Credentials

LEED AP Neighborhood Development, U.S. Green Building Council

Honors and Awards

- Dean's Honor List, Albers School of Business and Economics, Seattle University, Fall 2018.
- Best Graduate Design Award, Hebei University of Technology, 2008.
- Second Prize, National Undergraduate Urban Design Competition, Urban Planning Discipline Steering Committee, Ministry of Construction, China, 2007.